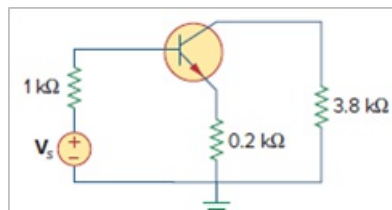


Problem

Calculate A_v , A_i , Z_{in} , and Z_{out} for the transistor network in Fig. 19.132. Assume that

$$\begin{aligned} h_{ie} &= 2 \text{ k}\Omega, & h_{re} &= 2.5 \times 10^{-4} \\ h_{fe} &= 150, & h_{oe} &= 10 \text{ }\mu\text{S} \end{aligned}$$

Figure 19.132



Step-by-step solution

Step 1 of 11

Consider the following circuit diagram:

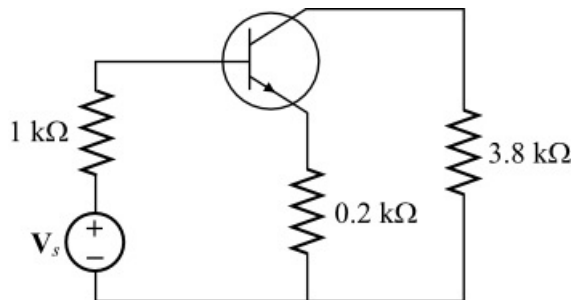


Figure 1

Step 2 of 11

Draw the equivalent circuit diagram of Figure 1.

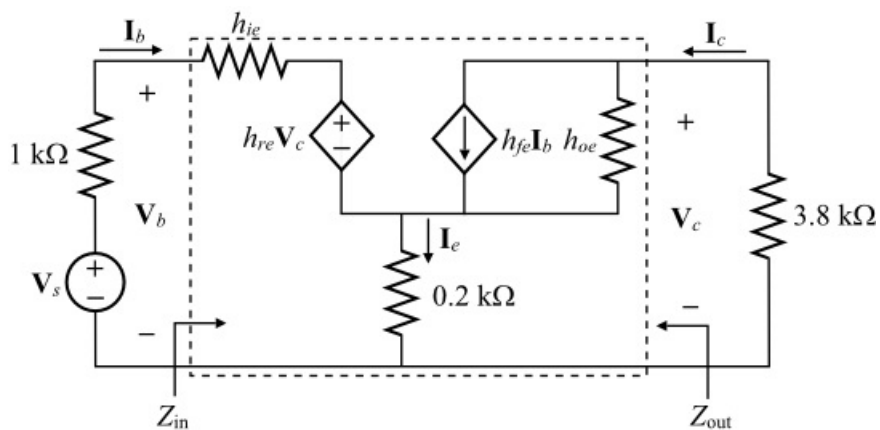


Figure 2

Step 3 of 11

From the figure 2, write the expression for $0.2 \text{ k}\Omega$ using Kirchhoff's current law.

$$\mathbf{I_e = I_b + I_c \dots\dots (1)}$$

Apply Kirchhoff's voltage law to left side loop.

$$\mathbf{V_b = h_{ie}I_b + h_{re}V_c + 200I_e}$$

$$\mathbf{V_b = h_{ie}I_b + h_{re}V_c + 200(I_b + I_c) \dots\dots (2)}$$

Apply Kirchhoff's current law at node V_c .

$$\mathbf{I_c = h_{fe}I_b + \frac{V_c}{200 + \frac{1}{h_{oe}}} \dots\dots (3)}$$

Step 4 of 11

Write the expression for output voltage.

$$V_c = -I_c R_L$$

Substitute $-I_c R_L$ for V_c in the equation (3).

$$I_c = h_{fe} I_b + \frac{-I_c R_L}{200 + \frac{1}{h_{oe}}}$$

$$\left(1 + \frac{R_L}{200 + \frac{1}{h_{oe}}} \right) I_c = h_{fe} I_b$$

$$\left(\frac{200 + \frac{1}{h_{oe}} + R_L}{200 + \frac{1}{h_{oe}}} \right) I_c = h_{fe} I_b$$

$$\left(\frac{1 + h_{oe}(R_L + 200)}{1 + h_{oe}200} \right) I_c = h_{fe} I_b$$

On further simplification:

$$\frac{I_c}{I_b} = \frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} \dots\dots (4)$$

$$A_i = \frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} \quad \left(\text{since } \frac{I_c}{I_b} = A_i \right)$$

$$= \frac{(150)[1 + (10 \times 10^{-6})(200)]}{1 + (10 \times 10^{-6})(3800 + 200)}$$

$$= \frac{(150)[1.002]}{1 + 0.04}$$

$$= \frac{150.3}{1.04}$$

$$= 144.5$$

Therefore, the current gain for the transistor circuit, A_i is 144.5.

From the equation (4),

$$I_c = \left(\frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} \right) I_b$$

Substitute $h_{fe} I_b + \frac{V_c}{240 + \frac{1}{h_{oe}}}$ for I_c in the above equation.

$$\left(\frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} \right) I_b = h_{fe} I_b + \frac{V_c}{200 + \frac{1}{h_{oe}}}$$

$$\left(\frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} - h_{fe} \right) I_b = \frac{V_c}{200 + \frac{1}{h_{oe}}}$$

$$I_b = \frac{V_c}{\left(200 + \frac{1}{h_{oe}} \right) \left(\frac{h_{fe}(1 + h_{oe}200)}{1 + h_{oe}(R_L + 200)} - h_{fe} \right)}$$

Substitute $\frac{V_c}{\left(200 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} - h_{fe}\right)}$ for I_b and $-\frac{V_c}{R_L}$ for I_c in the equation (2).

$$V_b = h_{ie}I_b + h_{re}V_c + 200(I_b + I_c)$$

$$V_b = (h_{ie} + 200) \left[\frac{V_c}{\left(200 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} - h_{fe}\right)} \right] + h_{re}V_c - \frac{200V_c}{R_L}$$

$$V_b = \left[\frac{h_{ie} + 200}{\left(200 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} - h_{fe}\right)} + h_{re} - \frac{200}{R_L} \right] V_c$$

$$\frac{V_b}{V_c} = \left[\frac{h_{ie} + 200}{\left(200 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} - h_{fe}\right)} + h_{re} - \frac{200}{R_L} \right]$$

Step 6 of 11

On Further simplification:

$$\begin{aligned} \frac{1}{A_v} &= \left[\frac{h_{ie} + 200}{\left(200 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} - h_{fe}\right)} + h_{re} - \frac{200}{R_L} \right] \\ &= \left[\frac{2000 + 200}{\left(200 + \frac{1}{10 \times 10^{-6}}\right)\left(\frac{(150)(1+(10 \times 10^{-6})(200))}{1+(10 \times 10^{-6})(3800+200)} - 150\right)} + 2.5 \times 10^{-4} - \frac{200}{3800} \right] \\ &= \left[\frac{2200}{(100200)\left(\frac{(150)(1.002)}{1.04} - 150\right)} + 2.5 \times 10^{-4} - 0.05263 \right] \\ &= \left[\frac{2200}{(100200)(144.5 - 150)} + 2.5 \times 10^{-4} - 0.05263 \right] \\ &= [-0.004 + 2.5 \times 10^{-4} - 0.05263] \\ &= -0.05638 \\ A_v &= \frac{1}{-0.05638} \\ &= -17.74 \end{aligned}$$

Step 7 of 11

Therefore, the voltage gain of the transistor circuit, A_v is -17.74.

Step 8 of 11

From the equation (4):

$$\frac{I_c}{I_b} = \frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)}$$

$$I_c = \frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} I_b$$

Substitute $-I_c R_L$ for V_c in the equation (2).

$$V_b = h_{ie} I_b + h_{re} (-I_c R_L) + 200(I_b + I_c)$$

$$= (h_{ie} + 200) I_b + (200 - h_{re} R_L) I_c$$

Now substitute $\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} I_b$ for I_c in the above equation.

$$V_b = (h_{ie} + 200) I_b + (200 - h_{re} R_L) \left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} \right) I_b$$

$$\frac{V_b}{I_b} = h_{ie} + 200 + (200 - h_{re} R_L) \left(\frac{h_{fe}(1+h_{oe}200)}{1+h_{oe}(R_L+200)} \right)$$

$$Z_{in} = h_{ie} + 200 + \left(\frac{h_{fe}(1+h_{oe}200)(200 - h_{re} R_L)}{1+h_{oe}(R_L+200)} \right)$$

Calculate the value of the input impedance.

$$Z_{in} = h_{ie} + 200 + \left(\frac{h_{fe}(1+h_{oe}200)(200 - h_{re} R_L)}{1+h_{oe}(R_L+200)} \right)$$

$$= 2000 + 200 + \left(\frac{(150)(1+(10 \times 10^{-6})(200))(200 - (2.5 \times 10^{-4})(3800))}{1+(10 \times 10^{-6})(3800+200)} \right)$$

$$= 2200 + \left(\frac{(150)(1.002)(200 - 0.95)}{1.04} \right)$$

$$= 2200 + \frac{(150)(1.002)(199.05)}{1.04}$$

$$= 2200 + 28766.725$$

$$= 30.966 \text{ k}\Omega$$

Therefore, the input impedance, Z_{in} is 30.966 kΩ.

Step 9 of 11

To calculate the output impedance, short the input source and replace load with 1 V voltage source. Draw the modified circuit diagram:

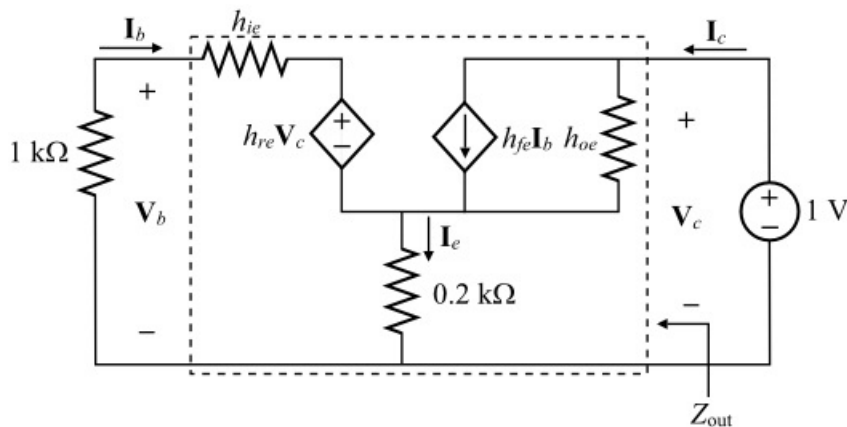


Figure 3

Step 10 of 11

Apply Kirchhoff's voltage law to left side loop.

$$(h_{ie} + 1000)I_b + h_{re}V_c + 200I_e = 0$$

But $V_c = 1 \text{ V}$

$$(h_{ie} + 1000)I_b + h_{re} + 200(I_b + I_c) = 0$$

$$(h_{ie} + 1000 + 200)I_b + h_{re} + 200I_c = 0 \dots\dots (5)$$

Apply Kirchhoff's current law at node V_c .

$$I_c = h_{fe}I_b + \frac{1}{200 + \frac{1}{h_{oe}}}$$

$$h_{fe}I_b = I_c - \frac{1}{200 + \frac{1}{h_{oe}}}$$

$$I_b = \frac{1}{h_{fe}} \left(I_c - \frac{1}{200 + \frac{1}{h_{oe}}} \right) \dots\dots (6)$$

Substitute $\frac{1}{h_{fe}} \left(\mathbf{I}_c - \frac{1}{200 + \frac{1}{h_{oe}}} \right)$ for $\mathbf{I}_b$ in the equation (5).

$$(h_{ie} + 1000 + 200) \left[\frac{1}{h_{fe}} \left(\mathbf{I}_c - \frac{1}{200 + \frac{1}{h_{oe}}} \right) \right] + h_{re} + 200\mathbf{I}_c = 0$$

$$\left(\frac{h_{ie} + 1000 + 200}{h_{fe}} + 200 \right) \mathbf{I}_c = \frac{h_{ie} + 1000 + 200}{h_{fe} \left(200 + \frac{1}{h_{oe}} \right)} - h_{re}$$

$$\mathbf{I}_c = \left(\frac{h_{fe}}{h_{ie} + 1000 + 200 + 200h_{fe}} \right) \left(\frac{h_{oe}(h_{ie} + 1000 + 200)}{h_{fe}(1 + 200h_{oe})} - h_{re} \right)$$

Calculate the output impedance.

$$\begin{aligned} Z_{out} &= \frac{1}{\mathbf{I}_c} \\ &= \frac{1}{\left(\frac{h_{fe}}{h_{ie} + 1000 + 200 + 200h_{fe}} \right) \left(\frac{h_{oe}(h_{ie} + 1000 + 200)}{h_{fe}(1 + 200h_{oe})} - h_{re} \right)} \\ &= \frac{2000 + 1000 + 200 + (200)(150)}{(150) \left(\frac{(10 \times 10^{-6})(2000 + 1000 + 200)}{(150)(1 + (200)(10 \times 10^{-6}))} - 2.5 \times 10^{-4} \right)} \\ &= \frac{2000 + 1000 + 200 + 30000}{(150) \left(\frac{0.032}{150.3} - 2.5 \times 10^{-4} \right)} \\ &= \frac{33200}{(150)(-3.709 \times 10^{-9})} \\ &= \frac{33200}{-5.56 \times 10^{-3}} \\ &= -5.96 \text{ M}\Omega \end{aligned}$$

Therefore, the output impedance, Z_{out} is $\boxed{-5.96 \text{ M}\Omega}$.